

10 Regression



10.2 Multiple Regression

From bivariate to multivariate linear regression — solving regression coefficients with orthogonal projection, the Gram matrix, and QR decomposition

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Master the algebraic structure of bivariate and multivariate linear regression models

2

Understand the regression plane and hyperplane geometrically as representations of variable relationships

3

The structure of the design matrix, including the constant column and each feature column

4

Understand multiple regression from the angle of orthogonal projection

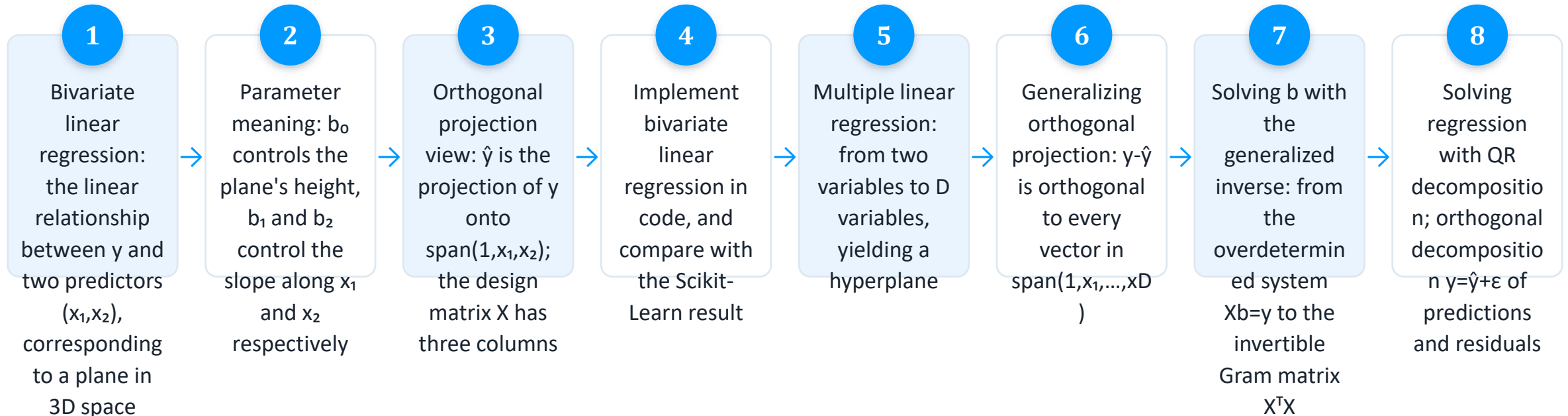
5

The principle of least-squares fitting

6

Use QR decomposition to solve regression problems

From the Plane of Bivariate Regression to the Hyperplane of Multiple Regression



Seven Angles for Reading Every Regression Topic in This Chapter

① Geometric view: the prediction is the orthogonal projection of the true-value vector onto the space spanned by the predictors

② Algebraic view: design matrix $\mathbf{X}$, Gram matrix $\mathbf{X}^T\mathbf{X}$, generalized inverse $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$

③ Optimization view: least squares — minimize the residual sum of squares $\|\boldsymbol{\epsilon}\|^2$

④ Statistical view: residuals, goodness of fit, significance testing — left for deeper coverage in Probability and Statistics Made Easy

⑤ Generalization path: from univariate to bivariate and multivariate linear regression, then to nonlinear polynomial regression

⑥ Coding view: hand-written matrix formulas and calling Scikit-Learn give identical results

⑦ Applications: prediction, trend fitting, and explaining variable relationships — among the most basic tools in data analysis

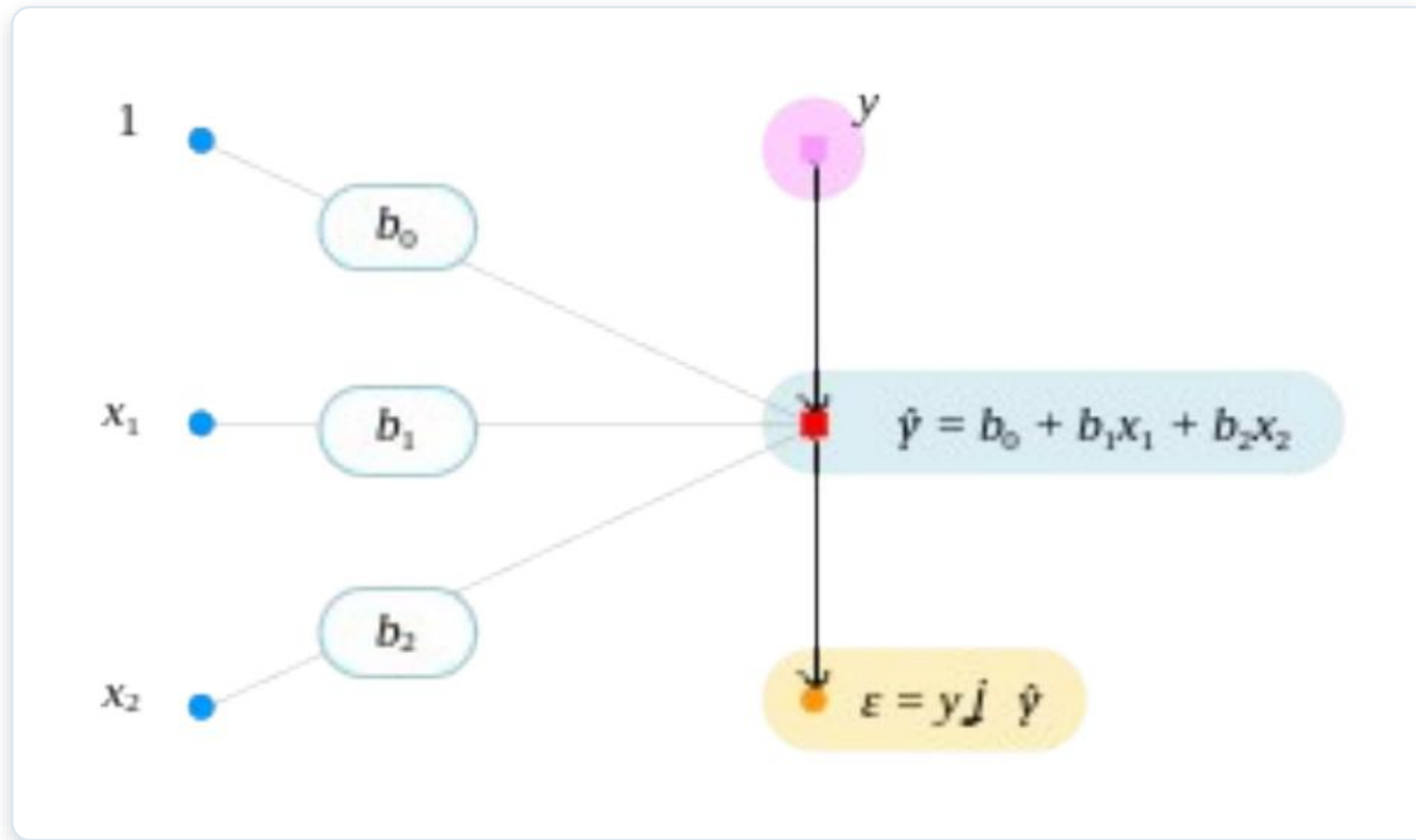
Bivariate Linear Regression Models a Linear Relationship Between y and Two Predictors (x_1, x_2)

$$y = b_0 + b_1x_1 + b_2x_2 + \varepsilon$$

- b_0 is the intercept, b_1 and b_2 are the predictor coefficients, ε is the residual
- Reading this section alongside 10.1 is strongly recommended — it lifts univariate regression to bivariate and multivariate

Figure 1 · Bivariate Linear Regression Variable Relationship

- Constant 1, predictors x_1 , x_2 multiply b_0 , b_1 , b_2 ; the sum is $\hat{y}$, and y minus $\hat{y}$ is the residual ε



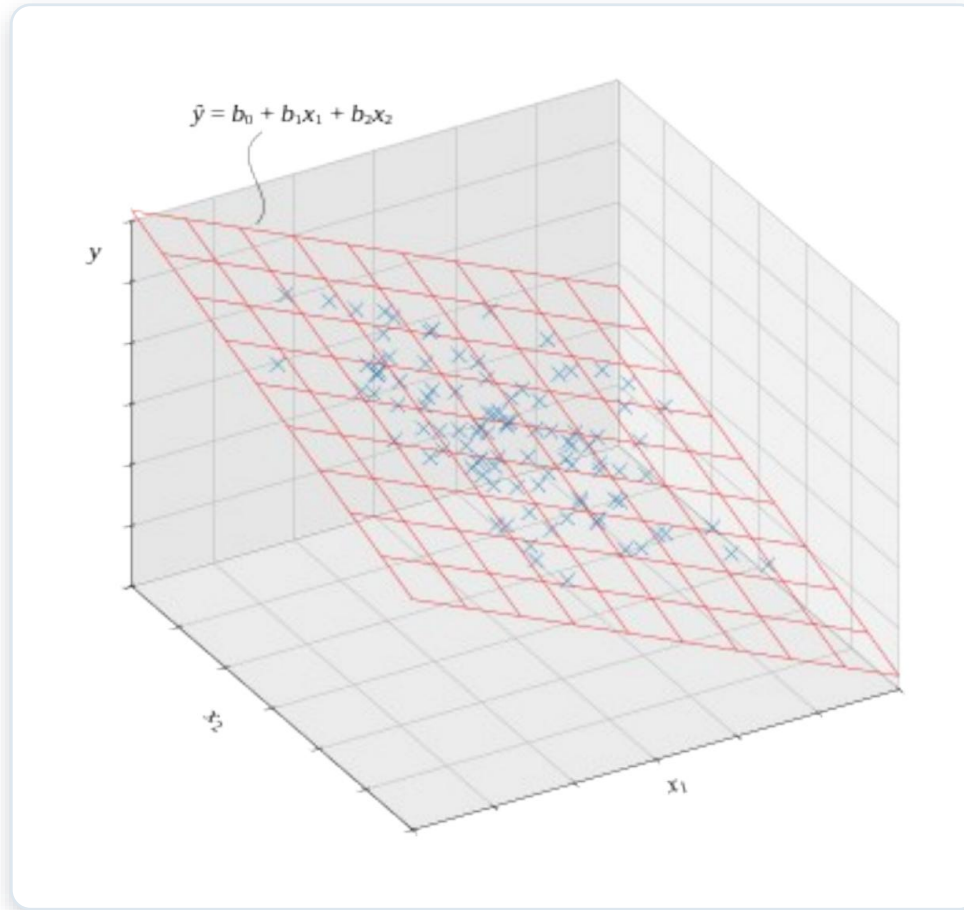
Geometrically, Bivariate Linear Regression Produces a Plane in 3D Space

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

- Bivariate linear regression finds the b_0 , b_1 , b_2 that make the plane fit the data scatter as closely as possible

Figure 2 · The Plane Corresponding to Bivariate Linear Regression

- The red mesh is the regression plane $\hat{y}=b_0+b_1x_1+b_2x_2$; the blue crosses are the training data scatter



How Do b_0 , b_1 , b_2 Affect the Plane $\hat{y}=b_0+b_1x_1+b_2x_2$?

b_0 The intercept — controls the plane's overall height, shifting it up or down without changing its tilt

b_1 Its magnitude controls the slope along x_1 ; its sign decides whether the plane rises or falls as x_1 increases

b_2 Its magnitude controls the slope along x_2 ; its sign decides whether the plane rises or falls as x_2 increases

Figure 3 · b_0 Controls the Overall Height of the Plane, $b_1=0$, $b_2=0$

- As b_0 grows, the plane shifts upward as a whole; the tilt direction stays unchanged

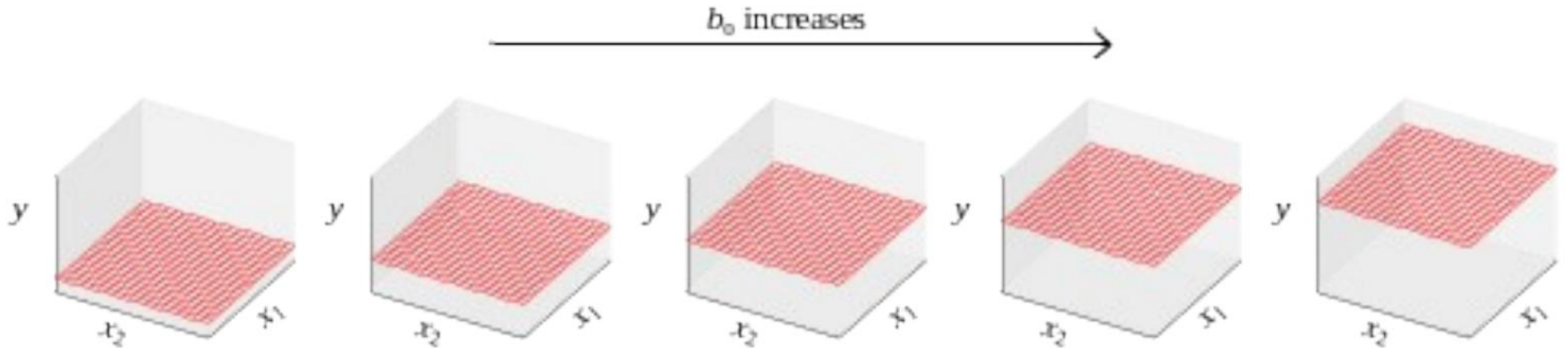


图 3. b_0 控制 $\hat{y} = b_0 + b_1x_1 + b_2x_2$ 平面的整体高度, $b_1 = 0$, $b_2 = 0$

Figure 4 · b_1 Controls the Slope Along x_1 , $b_0=0$, $b_2=0$

- As b_1 goes from negative to positive, the plane flips about the x_2 axis, reversing its slope

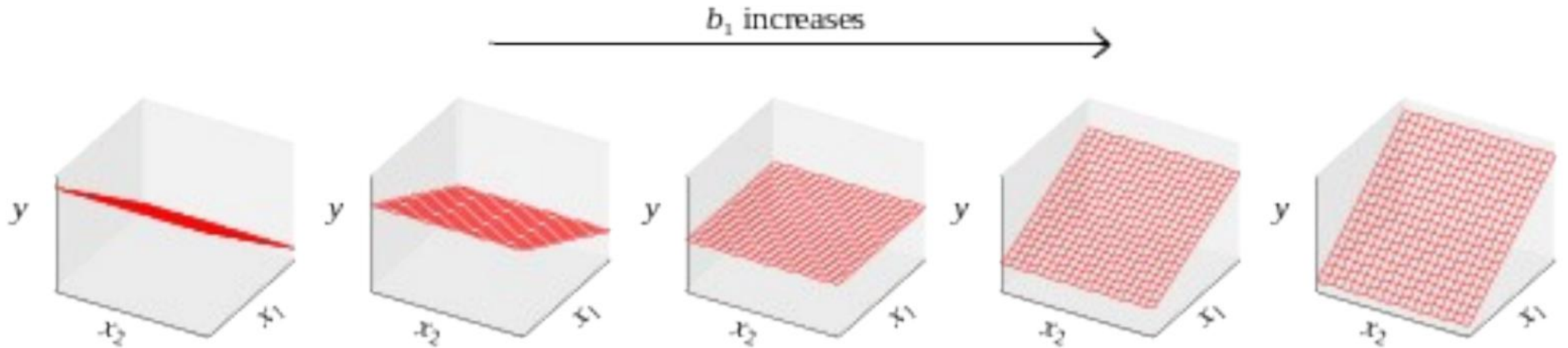


图 4. b_1 控制 $\hat{y} = b_0 + b_1x_1 + b_2x_2$ 平面沿 x_1 方向的斜率, $b_0 = 0$, $b_2 = 0$

Figure 5 · b_2 Controls the Slope Along x_2 , $b_0=0$, $b_1=0$

- As b_2 goes from negative to positive, the plane flips about the x_1 axis, reversing its slope

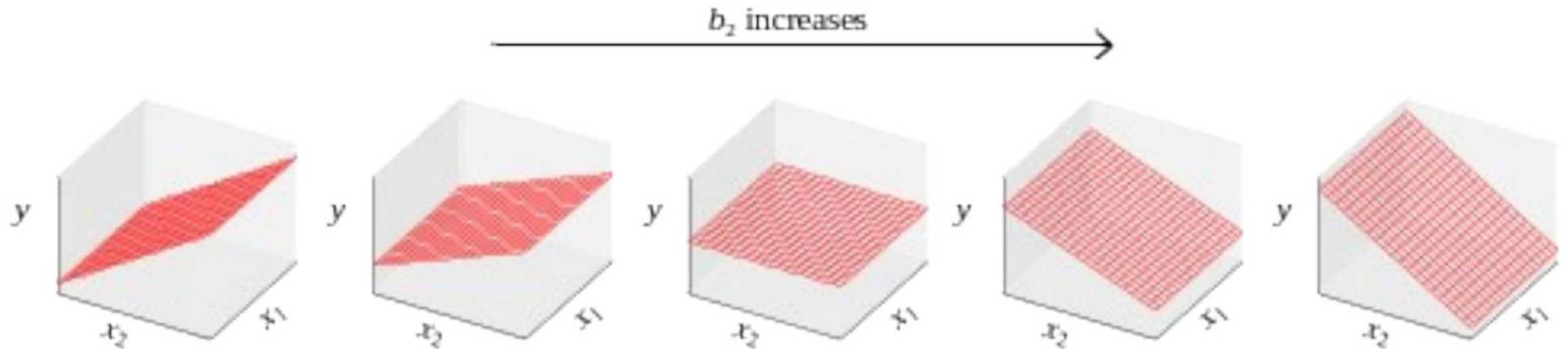


图 5. b_2 控制 $\hat{y} = b_0 + b_1x_1 + b_2x_2$ 平面沿 x_2 方向的斜率, $b_0 = 0$, $b_1 = 0$

Figure 6 · Three Planes Superimposed to Give $\hat{y}=b_0+b_1x_1+b_2x_2$

- $\hat{y}=b_0+b_1x_1+b_2x_2$ can be seen as the sum of three planes: a) $\hat{y}=b_0$; b) $\hat{y}=b_1x_1$; c) $\hat{y}=b_2x_2$

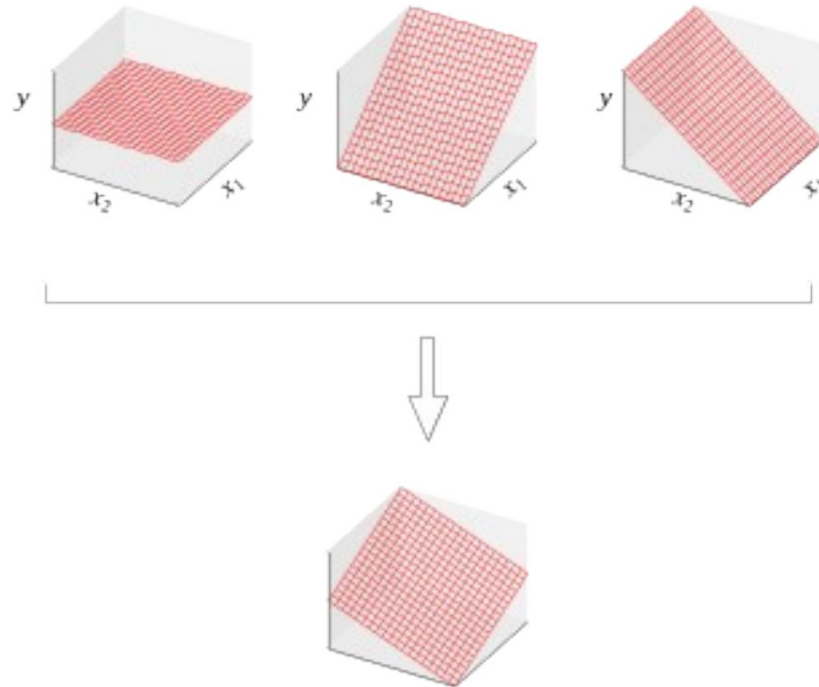


图 6. 三个平面叠加得到 $\hat{y} = b_0 + b_1x_1 + b_2x_2$

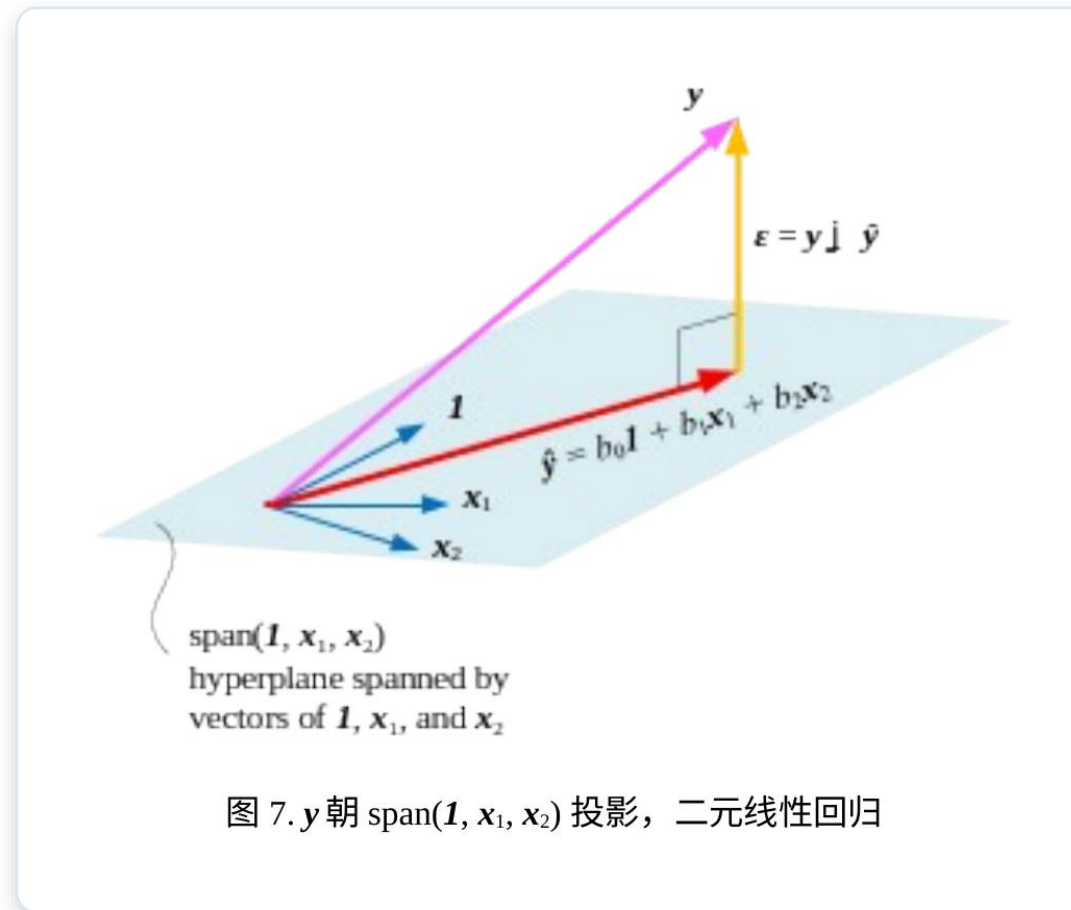
The True-Value Column Vector y Splits into a Prediction and a Residual

$$y = \hat{y} + \varepsilon = b_0 \mathbf{1} + b_1 \mathbf{x}_1 + b_2 \mathbf{x}_2 + \varepsilon$$

- $\hat{y}$ is the projection of y onto the plane $\text{span}(\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2)$ spanned by $\mathbf{1}$, $\mathbf{x}_1$, and $\mathbf{x}_2$

Figure 7 · y Projected Onto $\text{span}(1, x_1, x_2)$, Bivariate Linear Regression

- $\hat{y}$ is a linear combination of $1, x_1, x_2$; the error $\varepsilon = y - \hat{y}$ is orthogonal to this plane



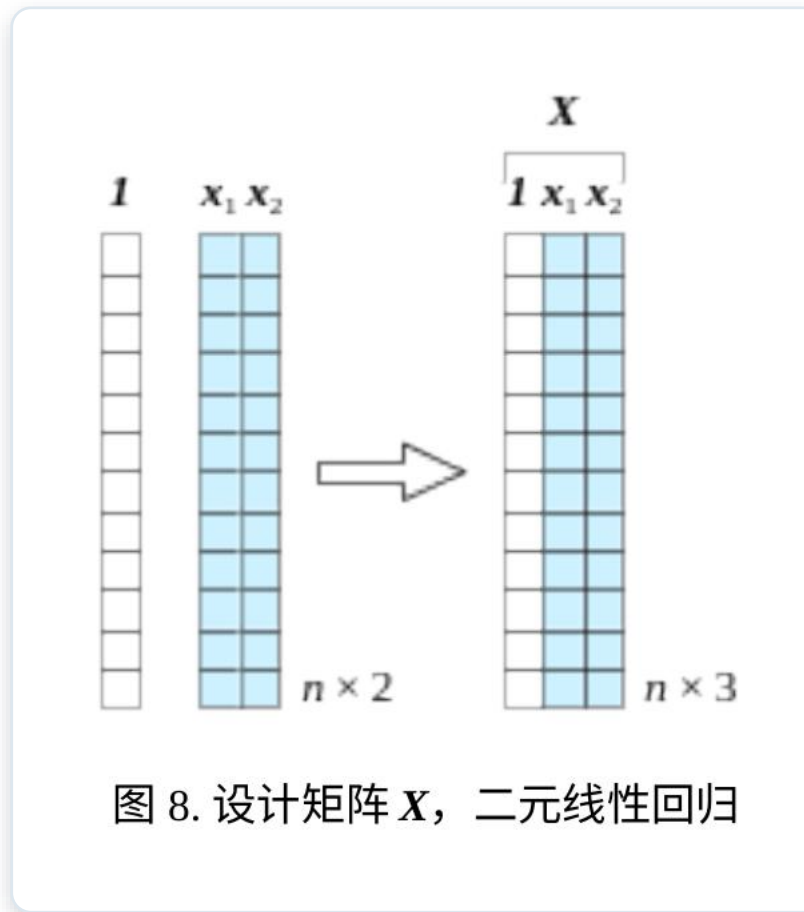
With Two Predictors, the Design Matrix X Has Three Column Vectors

$$X = [1 \ x_1 \ x_2]$$

- Assuming $1, x_1, x_2$ are linearly independent, X has full column rank, so the Gram matrix $X^T X$ has full rank and is invertible
- In practice the columns may be linearly dependent — such regressions need special treatment

Figure 8 · Design Matrix X , Bivariate Linear Regression

- Concatenate the all-ones column with the x_1 and x_2 data columns to get the $n \times 3$ design matrix X



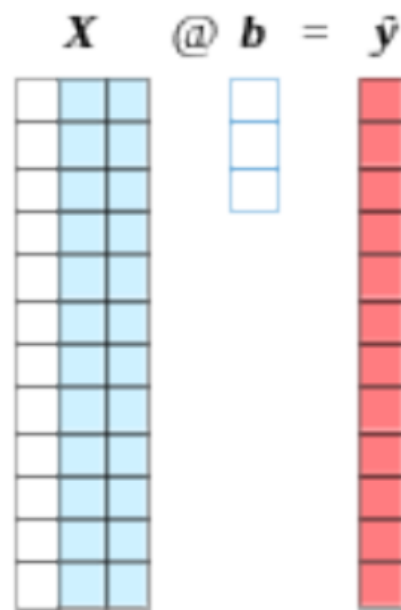
Writing the Linear Combination as Matrix Multiplication, $\hat{y} = Xb$

$$\hat{y} = [1 \ x_1 \ x_2] [b_0; b_1; b_2] = Xb$$

- b is the column vector stacking the three regression coefficients b_0, b_1, b_2

Figure 9 · Matrix Multiplication Produces $\hat{y}$, Bivariate Linear Regression

- Multiplying the design matrix X by the coefficient vector b directly gives the prediction vector $\hat{y}$

$$X @ b = \hat{y}$$


The diagram shows the matrix equation $X @ b = \hat{y}$. Matrix X is a 10x3 grid of light blue squares. Vector b is a 3x1 column of light blue squares. Vector $\hat{y}$ is a 10x1 column of red squares. The equation is centered above the respective visual representations.

图 9. 矩阵运算得到 $\hat{y}$ ，二元线性回归

Using the Gram Matrix and the Generalized Inverse to Solve for $\mathbf{b}$

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- $\mathbf{X}^T \mathbf{X}$ is the Gram matrix of $\mathbf{X}$; invert it and multiply by $\mathbf{X}^T$ to get the generalized inverse $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$
- Multiplying the generalized inverse $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ by $\mathbf{y}$ gives the optimal coefficient vector $\mathbf{b}$

Figure 10 · Computing $\mathbf{b}$, Bivariate Linear Regression

- $\mathbf{X}^T$ times $\mathbf{X}$ gives $\mathbf{X}^T\mathbf{X}$; its inverse times $\mathbf{X}^T$ is the generalized inverse; times $\mathbf{y}$ gives $\mathbf{b}$

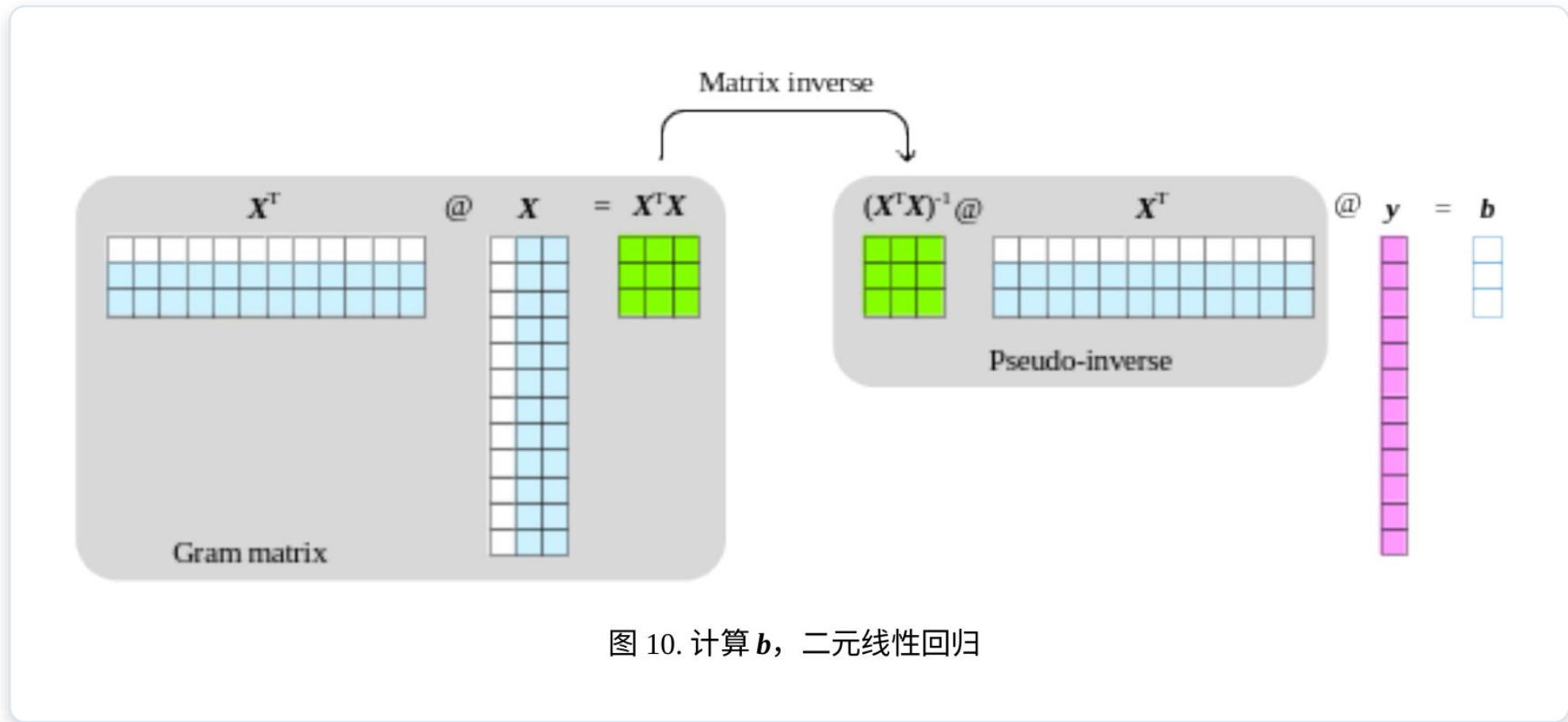


图 10. 计算 $\mathbf{b}$, 二元线性回归

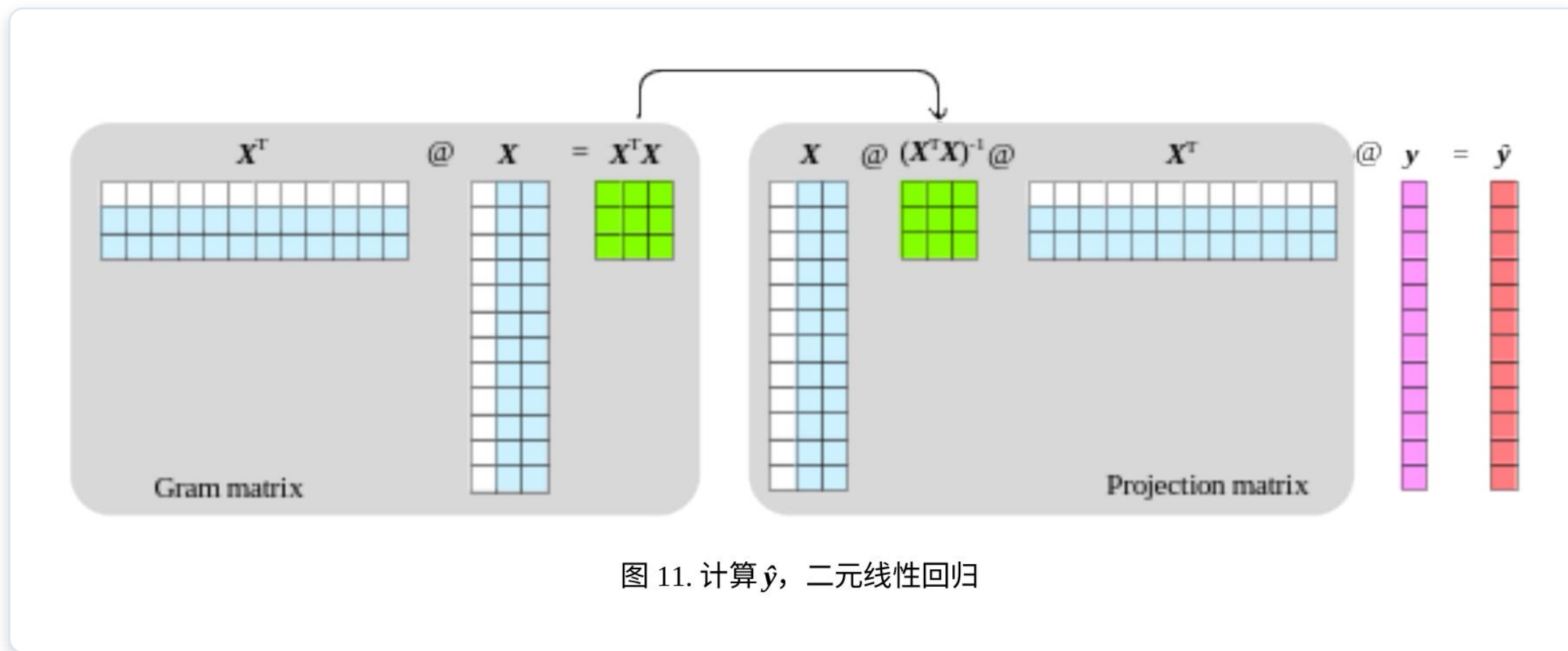
Substituting b 's Closed Form into $\hat{y}=Xb$ Reveals the Projection Matrix

$$\hat{y} = Xb = X(X^T X)^{-1} X^T y$$

- $X(X^T X)^{-1} X^T$ is the projection matrix that maps y onto the plane spanned by the columns of X

Figure 11 · Computing $\hat{y}$, Bivariate Linear Regression

- X , the inverse Gram matrix, and X^T multiply together into the projection matrix; times y gives $\hat{y}$



Solving Bivariate Linear Regression in Python

Hand-written matrix formula $b = \text{inv}(X.T@X)@X.T@y$ solves for the optimal parameters

- (a) $x_1, x_2 \sim \text{randn}$; y is a linear combination of x_1, x_2 plus noise
- (b) $X = \text{column_stack}(\text{ones}, x_1, x_2)$ — the $n \times 3$ design matrix
- (c) $b = \text{inv}(X^T X) @ X^T @ y$ — the standard least-squares formula
- (d) Build grid data X_{grid} , prepend a column of 1s for prediction
- (e) Predict on the grid: $y_{\text{pred}} = X_{\text{grid}} @ b$, corresponding to $\hat{y} = Xb$
- (f) Plot the 3D scatter of the training samples
- (g) Draw the regression plane as a wireframe over x_1, x_2

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 创建数据
np.random.seed(0)
n_samples = 100
x_12 = np.random.randn(n_samples, 2)
y = -3*x_12[:,0] + 2*x_12[:,1] + 1 + 0.25*np.random.randn(n_samples)

# 构造设计矩阵, 增加全1列向量
X = np.column_stack((np.ones_like(y), x_12))

## 创建线性回归模型并拟合数据
b = np.linalg.inv(X.T @ X) @ X.T @ y

## 拟合
# 生成回归平面的数据点
x1_grid, x2_grid = np.meshgrid(np.linspace(-3, 3, 10),
                                np.linspace(-3, 3, 10))
X_grid = np.column_stack((x1_grid.flatten(), x2_grid.flatten()))
X_grid = np.column_stack((np.ones_like(X_grid[:,0]), X_grid))

# 预测回归平面上的响应变量
y_pred = X_grid @ b
y_pred = y_pred.reshape(x1_grid.shape)

## 可视化
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
# 绘制三维样本散点
ax.scatter(x_12[:,0], x_12[:,1], y, marker = 'x')

# 绘制回归平面
ax.plot_wireframe(x1_grid, x2_grid, y_pred, colors = 'r')

ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_zlabel('$y$')
ax.set_xlim([-3,3]); ax.set_ylim([-3,3])
ax.set_zlim([-15,15])
ax.set_proj_type('ortho'); ax.view_init(azim=-120, elev=30)
```

Automating Model Training with Scikit-Learn

Study LA_10_02_02.ipynb on your own: it uses Scikit-Learn's LinearRegression tool to solve the same bivariate regression problem

The underlying math is identical to least squares, and the results exactly match LA_10_02_01.ipynb — compare the two

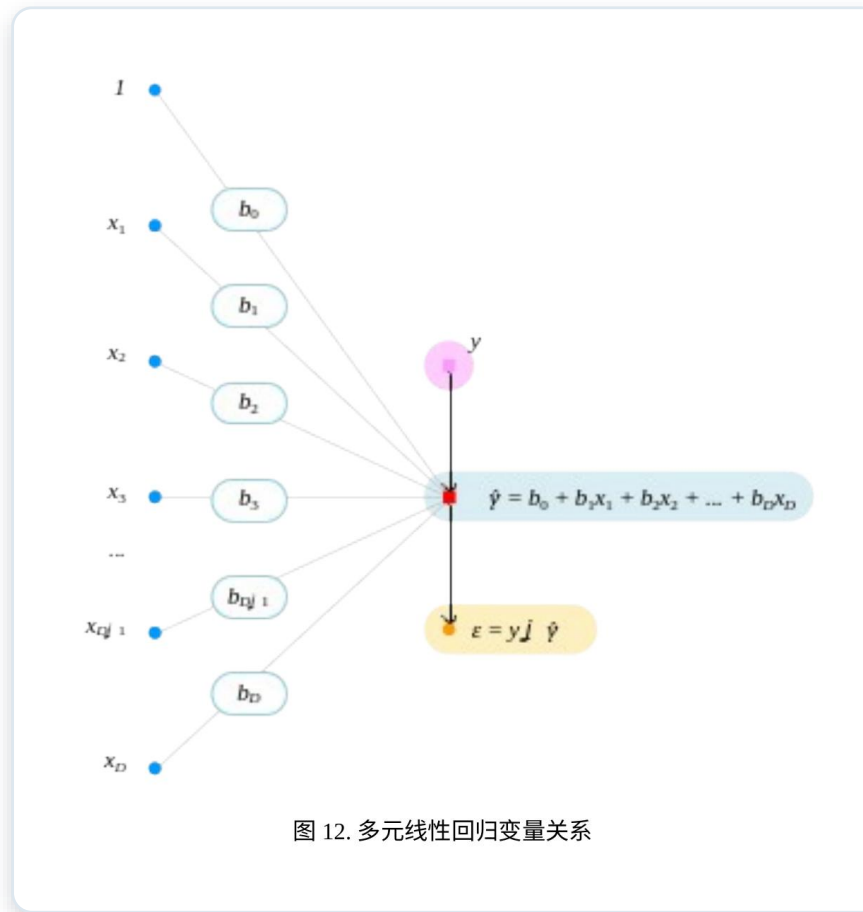
Multiple Linear Regression Models a Linear Relationship Between y and Several Predictors

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_Dx_D + \varepsilon$$

- b_0 is the intercept, $b_1, b_2, \dots, b_D$ are the predictor coefficients, ε is the residual, D is the number of predictors
- Geometrically, multiple linear regression produces a hyperplane — a generalization of a plane to higher dimensions

Figure 12 · Multiple Linear Regression Variable Relationship

- Constant 1 and x_1 through x_D multiply b_0 through b_D ; the sum is $\hat{y}$, and y minus $\hat{y}$ is ϵ



Viewed as Vector Spaces, $1, x_1, \dots, x_D$ Span a Hyperplane

$$y = \hat{y} + \varepsilon, \quad \hat{y} = b_0 \mathbf{1} + b_1 x_1 + \dots + b_D x_D$$

- Projecting y onto $\text{span}(1, x_1, x_2, \dots, x_D)$ gives $\hat{y}$ — meaning $\hat{y}$ is a linear combination of $1, x_1, \dots, x_D$

Figure 13 · y Projected Onto the Hyperplane $\text{span}(1, x_1, x_2, \dots, x_D)$, Multiple Linear Regression

- The error $y - \hat{y}$ is orthogonal to $\text{span}(1, x_1, \dots, x_D)$; $\text{span}(x_1, \dots, x_D)$ and $\text{span}(\varepsilon)$ are complements

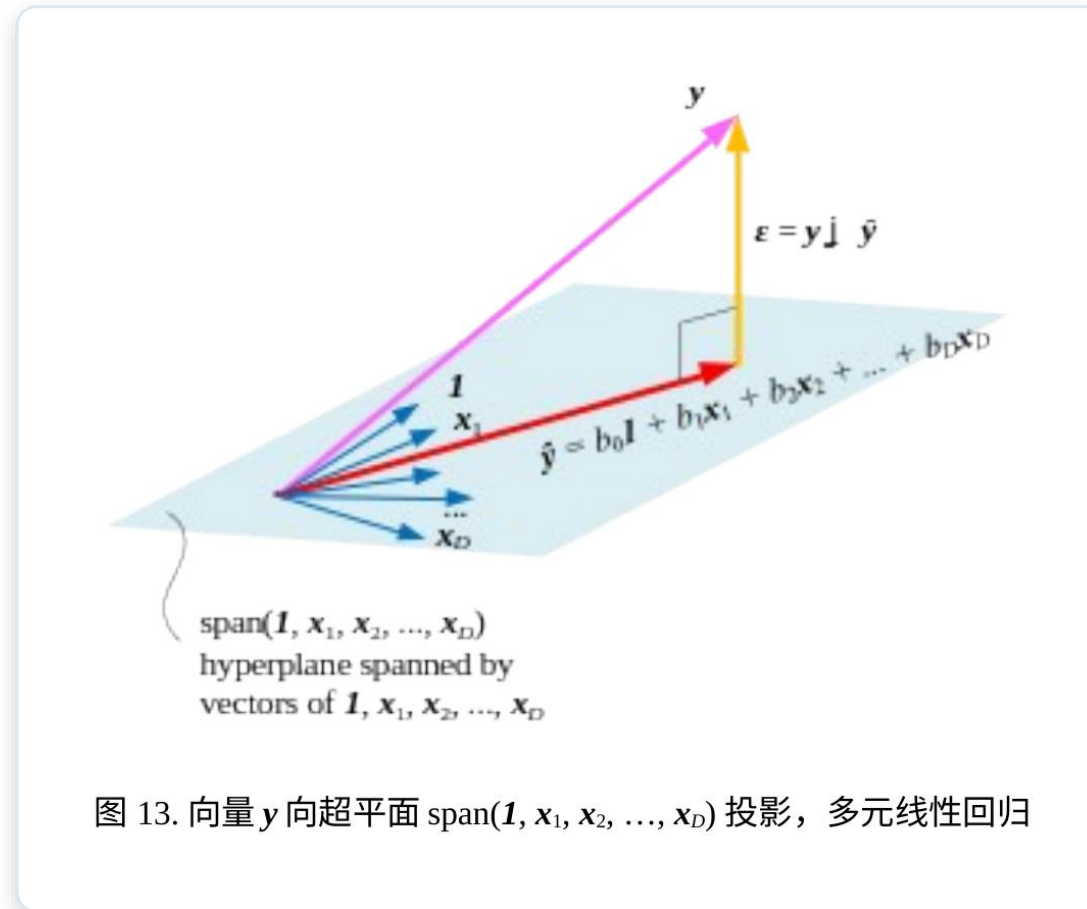


图 13. 向量 y 向超平面 $\text{span}(1, x_1, x_2, \dots, x_D)$ 投影，多元线性回归

The Error $y - \hat{y}$ Is Orthogonal to Each of $x_1, x_2, \dots, x_D$, Yielding the Design Matrix

$$X_{n \times (D+1)} = [1 \ x_1 \ x_2 \ \dots \ x_D]$$

- Orthogonality conditions $x_1^T(y - \hat{y}) = 0, \dots, x_D^T(y - \hat{y}) = 0$ combine into the matrix form $X^T(y - \hat{y}) = 0$

Figure 14 · Design Matrix X , Multiple Linear Regression

- Concatenate the all-ones column with x_1 through x_D to get the $n \times (D+1)$ design matrix X

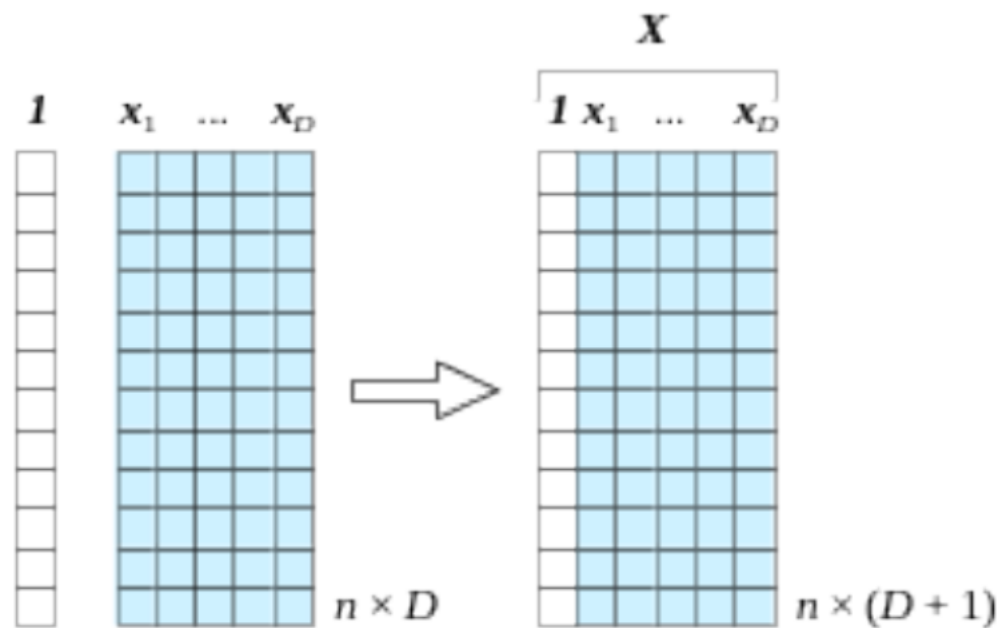


图 14. 设计矩阵 X , 多元线性回归

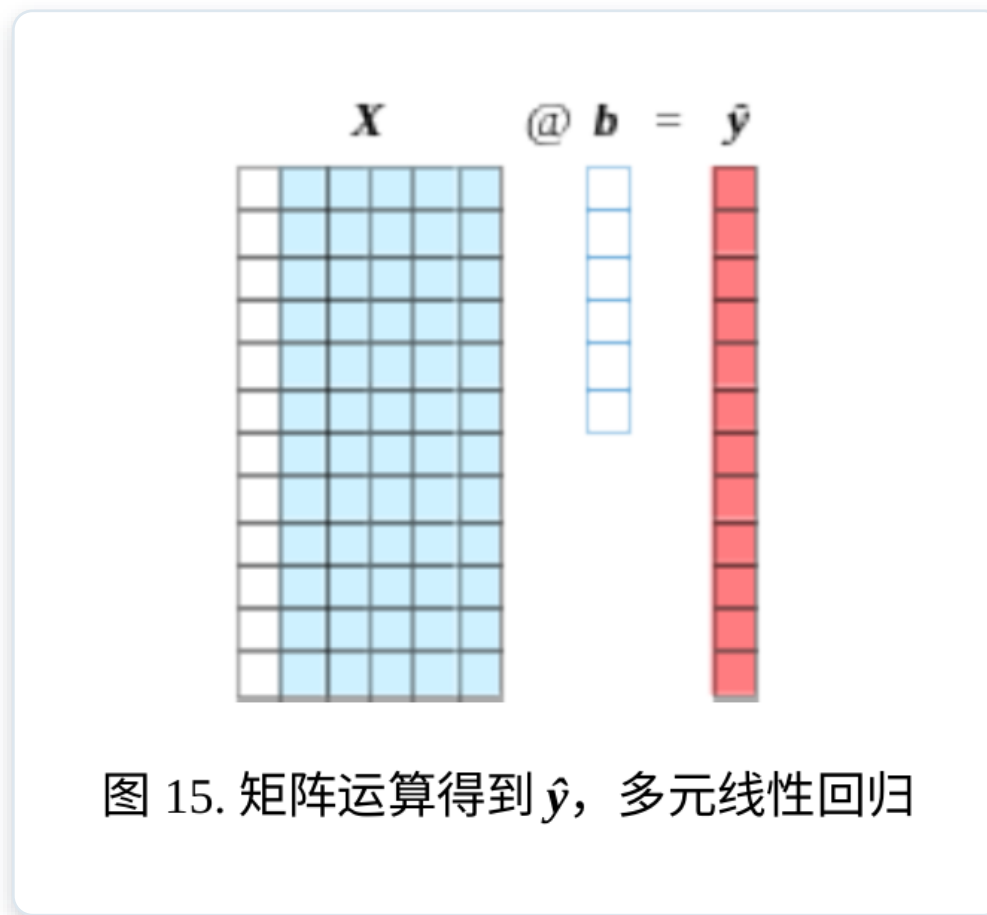
The Prediction Column Vector $\hat{y}$ Is Computed by Matrix Multiplication

$$\hat{y} = Xb$$

- b is the column vector stacking the $D+1$ regression coefficients $b_0, b_1, \dots, b_D$

Figure 15 · Matrix Multiplication Produces $\hat{y}$, Multiple Linear Regression

- Multiplying the design matrix X by the coefficient vector b directly gives the prediction vector $\hat{y}$



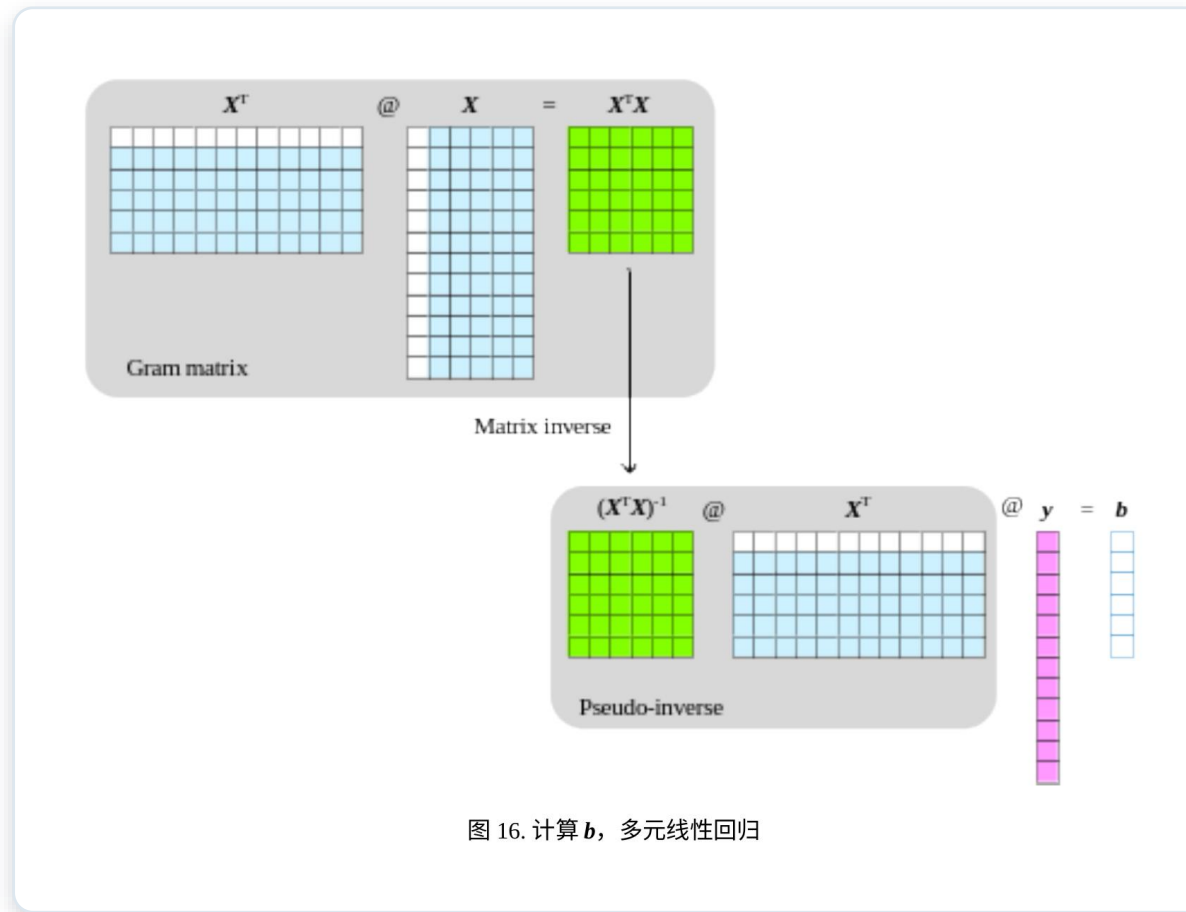
If the Gram Matrix Is Invertible, $\mathbf{b}$ Has the Closed Form

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- $\mathbf{X}^T \mathbf{X}$ is invertible only when the Gram matrix has full rank, which requires $\mathbf{X}$ to have full column rank

Figure 16 · Computing $\mathbf{b}$, Multiple Linear Regression

- $\mathbf{X}^T$ times $\mathbf{X}$ gives $\mathbf{X}^T\mathbf{X}$; its inverse times $\mathbf{X}^T$ is the generalized inverse; times $\mathbf{y}$ gives $\mathbf{b}$



Solving for $\mathbf{b}$ Is Essentially Solving an Overdetermined System

The overdetermined system $\mathbf{X}\mathbf{b}=\mathbf{y}$ has more equations (n samples) than unknowns ($D+1$ coefficients)

Left-multiplying both sides by $\mathbf{X}^T$ gives $\mathbf{X}^T\mathbf{X}\mathbf{b}=\mathbf{X}^T\mathbf{y}$, where $\mathbf{X}^T\mathbf{X}$ is the Gram matrix of $\mathbf{X}$

Left-multiplying by the inverse of $\mathbf{X}^T\mathbf{X}$ gives $\mathbf{b}=(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ — this is solving $\mathbf{b}$ from the overdetermined-system viewpoint

Performing a Full QR Decomposition on the Design Matrix X

$$X = QR, \quad Q = [Q_1 \ Q_2], \quad R = [R_1; \mathbf{0}]$$

- Q is a square orthogonal matrix; rearranging the block product gives the economy-size QR decomposition $X = Q_1 R_1$

R_1 Is Upper-Triangular, Which Makes Inversion Easier

The matrix product can be written $\hat{y} = Q_1 R_1 b$

Left-multiplying by the semi-orthogonal matrix Q_1^T gives $Q_1^T \hat{y} = R_1 b$,
since $Q_1^T Q_1 = I$

R_1 is upper-triangular, so its inverse is easier to compute than that of the Gram matrix — this is the advantage of QR decomposition for solving regression

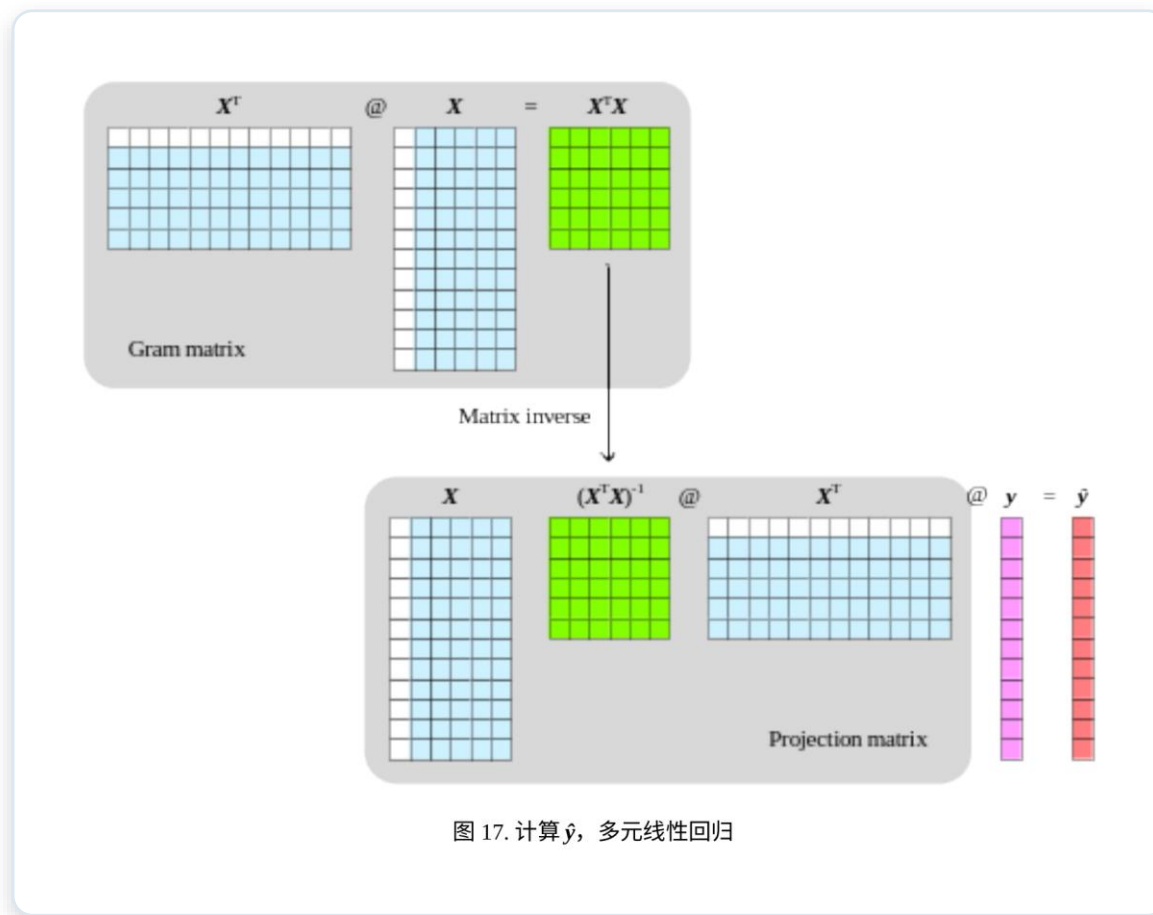
The Fitted Vector $\hat{y}$ Can Be Written as a Projection Matrix Times y

$$\hat{y} = Xb = X(X^T X)^{-1} X^T y$$

- $X(X^T X)^{-1} X^T$ is the projection matrix mapping y onto the hyperplane spanned by the columns of X — earlier called the hat matrix

Figure 17 · Computing $\hat{\mathbf{y}}$, Multiple Linear Regression

- X , the inverse Gram matrix, and X^T multiply together into the projection matrix; times $\mathbf{y}$ gives $\hat{\mathbf{y}}$



The Residual Vector ϵ Can Also Be Written as a Projection Matrix Times y

From $\epsilon = y - \hat{y}$, the residual column vector can be written $\epsilon = (I - X(X^T X)^{-1} X^T)y$

$\text{span}(\epsilon)$ is the orthogonal complement of $\text{span}(1, x_1, \dots, x_D)$; viewed another way, ϵ is the normal vector of that hyperplane

$I - X(X^T X)^{-1} X^T$ is also a projection matrix, mapping y into the "residual" space to give ϵ

Adding $\hat{y}$ and ϵ : $\hat{y} + \epsilon = y$ — the vector y decomposes orthogonally into the prediction $\hat{y}$ and the error ϵ

SUMMARY

What Does Moving from Bivariate to Multiple Regression Reveal?

Bivariate linear regression corresponds to a plane in 3D space;
multiple linear regression corresponds to a hyperplane

b_0 controls the plane's height; $b_1, \dots, b_D$ control the slope along each predictor's direction

The prediction $\hat{y}$ is the orthogonal projection of y onto $\text{span}(1, x_1, \dots, x_D)$; the residual ϵ is orthogonal to that space

With design matrix $X = [1, x_1, \dots, x_D]$ and an invertible Gram matrix $X^T X$,
 $b = (X^T X)^{-1} X^T y$

The projection matrix $X(X^T X)^{-1} X^T$ maps y to $\hat{y}$; $I - X(X^T X)^{-1} X^T$ maps y to ϵ
— the two are orthogonal complements

QR decomposition writes $X = Q_1 R_1$ and inverts the upper-triangular R_1
— another route to solving regression

Key Terms in This Section

Bivariate / Multiple Linear Regression 多元线性回归 —— a model in which the dependent variable varies linearly with two or more predictors

Intercept, Slope Coefficients 截距、斜率系数 —— b_0 (intercept) and $b_1, \dots, b_D$ (slope coefficients)

Design Matrix 设计矩阵 X —— its first column is all ones, the rest are the predictor values

Generalized Inverse 广义逆 / 伪逆 —— $(X^T X)^{-1} X^T$

QR Decomposition QR分解 —— $X=QR$, solving regression with an orthogonal matrix and an upper-triangular matrix

Hyperplane 超平面 —— the generalization of a plane in 3D space to higher dimensions

Residual 残差 —— the difference between the true and predicted value, $\varepsilon = y - \hat{y}$

Gram Matrix 格拉姆矩阵 —— $X^T X$, used to solve for the optimal parameters in regression

Overdetermined System 超定方程组 —— a system $Xb=y$ with more equations than unknowns

Projection Matrix, Hat Matrix 投影矩阵、帽子矩阵 $H = X(X^T X)^{-1} X^T$, satisfying $\hat{y} = Hy$

FURTHER STUDY

Suggested Further Reading

Review · 10.1 From Projection to Regression

This section builds on univariate linear regression and orthogonal projection — read them in parallel

Further Reading · 10.3 Polynomial Regression

The next section shows how nonlinear relationships become linear regression problems

Further Reading · Probability and Statistics Made Easy

Significance testing, confidence intervals, goodness of fit, multicollinearity

Open-Source Repository

Code and figures for the entire book: <https://github.com/Visualize-ML>

Questions Q1 – Q9

Q1 Given code that generates multiple-regression predictor and response data, compute b , $\hat{y}$, and ε in Python.

Q3 Why does bivariate linear regression correspond to a plane in 3D space rather than a line?

Q5 Why must the first column of the design matrix X be all ones? What happens to the model if it is removed?

Q7 Prove that the residual vector ε is orthogonal to every column of the design matrix X .

Q9 In bivariate linear regression, what shape does the regression plane degenerate into when $b_1=0$ or $b_2=0$?

Q2 Solve the multiple linear regression in Q1 with `sklearn.linear_model`'s `LinearRegression`, and compare results.

Q4 In $\hat{y}=b_0+b_1x_1+b_2x_2$, which point must the plane pass through when $b_0=0$? Why?

Q6 Under what condition is the Gram matrix $X^T X$ not invertible? Give an example with linearly dependent predictors.

Q8 Does the projection matrix $H=X(X^T X)^{-1}X^T$ satisfy $H^2=H$? Verify with matrix operations.

Questions Q10 – Q17

Q10 Use numpy to build the design matrix X for a 5-variable linear regression, and compute its Gram matrix $X^T X$.

Q12 Use `numpy.linalg.qr` to compute an economy-size QR decomposition of X , and solve for b using Q_1 and R_1 .

Q14 Write a Python function `hat_matrix(X)` that returns the projection matrix $H = X(X^T X)^{-1} X^T$.

Q16 Use matplotlib to visualize the residual-sum-of-squares contour as two coefficients of a trivariate regression vary.

Q11 Write a Python function `fit_ols(X, y)` that returns the least-squares coefficient vector b .

Q13 Compare solving b by inverting the Gram matrix versus by QR decomposition — do the two methods agree?

Q15 Verify whether the residual vector $\varepsilon = (I - H)y$ and the prediction vector $\hat{y} = Hy$ are orthogonal (is their inner product 0)?

Q17 Write a Python function `r_squared(y, y_hat)` that computes the goodness-of-fit R^2 .

Questions Q18 – Q25

Q18 Explore: when the number of predictors D far exceeds the sample size n , is $X^T X$ still invertible? What problems arise?

Q20 Write a Python function `standardize(X)` that standardizes each column of the design matrix, then compare the resulting coefficients.

Q22 Verify in code whether $Q_1^T Q_1$ equals the identity matrix I , and explain its role in solving regression via QR decomposition.

Q24 Explore multiple linear regression on a real dataset (e.g. housing prices): pick 2-3 features, fit a model, and visualize it.

Q19 Generate a set of predictor data with multicollinearity, and observe how the condition number of $X^T X$ changes.

Q21 Explore: remove one predictor column from the design matrix, refit, and compare the change in R^2 and the coefficients.

Q23 Comprehensive exercise: write a Python function `predict(X_new, b)` that batch-predicts on new sample data.

Q25 Preview: the next section covers polynomial regression (10.3) — how does its design matrix relate to and differ from this section's?

10 Prompts to Explore with AI

- 1 Explain why bivariate linear regression is geometrically a plane, while multiple linear regression is a hyperplane.
- 2 Explain the role of the Gram matrix $X^T X$ in least-squares regression, and why it must have full rank to be invertible.
- 3 Use Python to demonstrate implementing multiple linear regression's least-squares solution from scratch with numpy.
- 4 Explain the relationship between the generalized inverse $(X^T X)^{-1} X^T$ and the projection matrix $X(X^T X)^{-1} X^T$.
- 5 Explain the numerical advantage of solving regression with QR decomposition over directly inverting the Gram matrix.

10 Prompts to Explore with AI

6 Explain why the residual vector ϵ is orthogonal to the column space of the design matrix X , and how this is used to solve for b .

7 Explain the origin of the name "hat matrix," and the geometric meaning of it satisfying $H^2=H$.

8 Explain how multicollinearity affects the estimation of multiple regression coefficients, and common ways to address it.

9 Explain the mathematical relationship between the residual sum of squares $\|\epsilon\|^2$ and the goodness-of-fit R^2 .

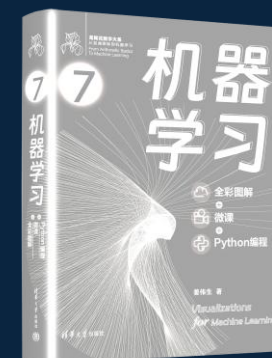
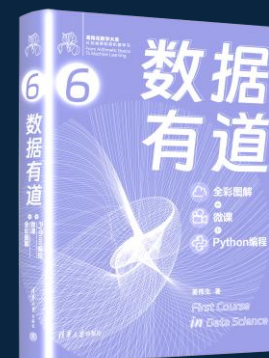
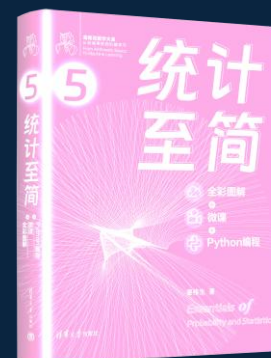
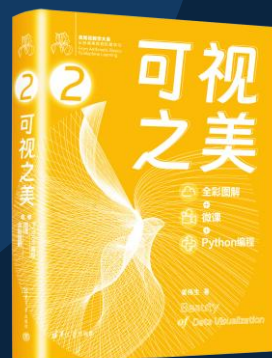
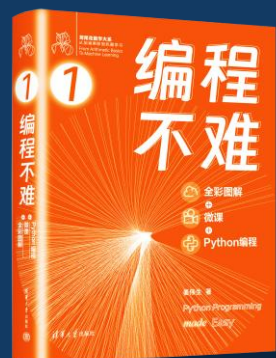
10 Use Python to visualize the same dataset fit with univariate, bivariate, and multiple linear regression, compared side by side.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

